

Integrating Copy Number Alterations and Transcriptomics with Machine Learning Improves
Survival Prediction in HPV-negative Head and Neck Cancer

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Table of Contents

Cover Page	1
Table of Contents	2
Acknowledgement of Major Assistance Received	3
Introduction	4
Methods	5
Results	7
Discussion and Conclusions	9
References	11
Appendix	14

Acknowledgement of Major Assistance Received

I received major assistance from Pranav Narnur, a Ph.D. student at Boston University. He aided in the conceptualization of the idea, acquisition of data, iterative testing, biological analysis, writing of the proposal, and development of the final research paper. I am grateful for his willingness to work with and teach me and his availability.

Introduction

Head and Neck Squamous Cell Carcinoma (HNSCC) is a type of cancer affecting the upper digestive tract, manifesting in regions such as the lips, the mouth, the larynx, the pharynx, and the nasal passages.¹ It ranks seventh in worldwide prevalence amongst all cancers, with over 900,000 new cases annually.² Two distinct subtypes have been identified—HPV-positive and HPV-negative HNSCC. The HPV-negative subtype, driven by exposure to carcinogens like alcohol and tobacco, is characterized by a significantly worse prognosis than the HPV-positive subtype, with the 5-year overall survival rate in HPV-unrelated cases standing at just 48%.^{3,4,28}

Because of its anatomical location, HPV-negative HNSCC causes immense patient suffering, stifling basic functions such as speech, breathing, and swallowing.⁵ The existing gold standard of treatment involves clinicians developing comprehensive chemoradiotherapeutic plans, integrating intensity-modulated radiation therapy and drugs such as cisplatin and pembrolizumab.⁶ Although a general combination of pharmacological and radiation-based treatment is key, many patients do not respond to clinical intervention.^{7,8} Therefore, determining what therapeutics to administer to a particular patient requires careful molecular characterization of tumors and subsequent assessment of patient prognosis.

Present clinical workflows involve histopathological staging of tumors, grading them from I-IV, with higher grades indicating a more advanced condition with poorer prognosis.⁹ However, this staging does not capture the full complexity of the disease at the molecular level, and patients with the same tumor stage can significantly diverge in their overall outcomes.¹⁰ To aid clinicians in making better treatment decisions, many computational workflows for predicting overall survival have been developed.

These approaches utilize a variety of datasets. The Cancer Genome Atlas (TCGA) and the Gene Expression Omnibus (GEO) repositories are the most frequently used, along with independent cohorts from medical centers throughout the world.^{11,12,13} From these repositories, clinical and/or genomic data are used to train and test predictive models. Many existing models harness only clinical variables like age, sex, and smoking status.^{14,15} However, with the increasing availability of next-generation sequencing methods, genomic modalities, such as transcriptomics and proteomics, are becoming more common.^{16,17} Despite the growing abundance of data, most survival papers do not incorporate data from more than one modality.^{18,19} This is an issue because HPV-negative HNSCC is known for its high mutational burden and complex biochemical landscape across all -omics layers. To accurately predict survival, multiple levels of analysis are important.²⁰

After clinical and/or -omics data are downloaded, preprocessed, and filtered, features most associated with overall survival must be selected, as including every feature would lead to extreme overfitting, reducing generalizability.²¹ However, selecting too few features prevents the model from learning the complex patterns associated with HPV-negative HNSCC.²² Thus, balanced feature count is necessary for adequate prediction of survival.

Once features have been chosen, specific machine learning architectures, such as DeepSurv, XGBoost, and CoxPH, are trained and tested.^{23,24} Traditionally, the model learns patterns in data on a “training” set and is evaluated on the test set using certain metrics, such as the concordance index (c-index), which assesses ability to order pairs of patients based on survival time.²⁵ The lack of external validation, a common issue, can upwardly skew this index, making it difficult to assess a model.²⁶ Some well-validated studies include Yin et al (2024), which incorporated immune markers and RNA-seq data into a predictor, achieving a c-index of

0.61 on a GEO dataset, and Zheng et al (2021), which developed a 7-gene prognostic signature and achieved an index of 0.641 on a different dataset.^{19, 27} When tested on an external cohort, c-indices rarely exceed 0.65.¹⁸

This project aims to: i. integrate multiple types of genomic data with clinical data for the development of an HPV-negative HNSCC predictive model; ii. externally validate the final model via the recently released clinical proteomic tumor analysis consortium (CPTAC); iii. illuminate the biological mechanisms underlying HPV-independent tumor progression.

Methods

Data Selection and Preprocessing

For the training dataset, The Cancer Genome Atlas (TCGA) RNA-sequencing (RNA-seq) and somatic copy number alteration (SCNA) datasets for HNSCC were downloaded from the National Cancer Institute's Genomic Data Commons Portal.³⁰ For external validation, The Clinical Proteome Tumor Analysis Consortium (CPTAC) RNA-seq and SCNA datasets for HNSCC were downloaded from this same portal.³¹ The TCGA and CPTAC are separate, well-annotated cohorts of cancer patients.

Sample-level filtering was performed using the tidyverse libraries in the R programming language. After filtering, the data comprised only HPV-negative HNSCC primary tumor samples. Patients with missing information for either survival time or censoring status were removed. The final RNA-seq and SCNA discovery datasets, sourced from the TCGA, each contained 390 samples. The final RNA-seq and SCNA validation datasets, sourced from the CPTAC, each contained 104 samples.

Feature-level filtering was then performed via the tidyverse ecosystem, shrinking the datasets to only contain autosomal protein coding genes. Additionally, only features present in both the discovery and validation datasets were kept. The filtered RNA-seq discovery and validation datasets both contained 17501 genes, and the final SCNA discovery and validation datasets each contained 4314 genes.

The RNA-seq data was DESeq2 normalized and log2-transformed, a standard step when working with RNA-seq data to account for biases in sequencing depth and library size.³²

Iterative Model Selection, Training, and Testing

Seven iterations of model training and testing were performed to find the combination of feature selection, model architecture, and hyperparameters which maximized test set performance. Three different models were trained for each architecture: a clinical-feature only model, an -omics only model, and a combined model. The trained models were assessed on the CPTAC validation dataset using the Harrell's concordance index (C-index), a goodness of fit measure employed for evaluating survival models.³³ The model with the highest C-index on the validation dataset was reported. The performances of the high-scoring models were compared to those of previously published HNSCC overall survival frameworks. For every model architecture, the following train/test splits were employed: 80% of the TCGA samples would be used for training (312 samples), 20% of TCGA samples would be used for cross-validation (78 samples), and 100% of the CPTAC samples would be used for testing.

The first and second training runs used base CoxPH and CoxNet models with approximately 100 features, commonly used models for survival prediction.³⁴ The third run

tested the random survival forests (RSF) architecture using a stability selection approach for feature selection. The third run resulted in a marked increase in test set C-Index, likely due to the RSF model avoiding many of the distributional assumptions other models make. The fourth run tried the gradient boosted survival (GBS) model architecture^{29,34}. The 5th run added the RidgeCox³⁴ model architecture. The 6th run modified the feature selection strategy across model architectures to implement an aggressive feature selection strategy, trying 15, 30, 50, 75, and 100 features.

The 7th and final training run implemented a block-constrained univariate feature selection, which would select the features most associated with survival in two separate blocks, the RNA-seq block and the SCNA block. This would ensure that both RNA-seq and SCNA features were present in the model. The formula used to select features was the C-Index deviation:

$$\text{Deviation}_j = |C\text{-index}_j - 0.5|$$

This measures the C-Index (ranging from 0.5, or random guessing, to 1, or perfect prediction) for each individual feature and subtracts it from 0.5. In other words, it measures how much better each feature predicts patient survival than random chance. Following this procedure, 70 RNA-seq features and 4 SCNA features were selected, and this unequal splitting proved to be optimal because, in general, the RNA-seq features with the highest C-Index deviations were more associated with survival than the SCNA features with the highest C-Index deviations. With that being said, the inclusion of the 4 SCNA features did prove to meaningfully improve prediction accuracy. For the RSF model, the following hyperparameters were tuned: `n_estimators`, `max_depth`, `min_samples_split`, `min_samples_leaf`, `max_features`, and `oob_score`.

Applying trained model to stratify patients into risk groups

Using the winning RSF model, patients in the CPTAC cohort were assigned a continuous risk score using the trained Random Survival Forest model's `predict()` function. This risk score represents the predicted cumulative hazard function (CHF) at the patient's observed follow-up time, and higher values indicate worse prognosis. The median risk score in the training set was used as a cutoff to classify patients into two broad prognostic groups: "higher risk" or "lower risk". This fixed cutoff was applied to the test set to classify patients in the CPTAC cohort. Then, two Kaplan Meier curves were generated for the CPTAC cohort, one for the higher risk and one for the lower risk group. Afterwards, a log-rank was performed to compare survival distributions between higher-risk and lower-risk groups. This test assesses the null hypothesis that survival curves are identical across groups.

Pathway enrichment analysis on prognostic signature

To characterize the biological processes captured by the 74-gene prognostic signature (70 RNA + 4 SCNA), an over-representation analysis (ORA) was performed using Enrichr via GSEAPy v1.1.11.^{35,36} The 74 unique gene symbols were queried against six pathway databases: GO Biological Process (2023), GO Molecular Function (2023), GO Cellular Component (2023), Reactome (2022), MSigDB Hallmark (2020), WikiPathways (2023), and KEGG (2021). An FDR-adjusted p-value cut-off of 0.25 was used to select significant pathway enrichments. The

top enriched terms were visualized as a dot plot, where the x-axis represents gene ratio (number of overlapping genes / gene set size), dot size reflects gene count, and dot color indicates FDR-adjusted p-value.

Results

Figure 1 lists the train and test set C-indices for the best-performing model from each of the 5 survival model architectures which were tested. The RSF model had train C-indices of 0.654 for the clinical-only model, 0.794 for the -omics only model, and 0.799 for the combined model. Importantly, the RSF combined model had the highest test C-index out of all models evaluated, at 0.664. The combined RSF model had an improved test C-index when compared with the clinical-only RSF (test C-index = 0.538) and the -omics only RSF model (test C-index = 0.626), demonstrating that integrating both clinical and -omics data improves survival prediction.



Figure 1: Performance of train and test set C-indices across survival models. 3 types of survival models (clinical-only, -omics-only, and combined) were evaluated across 5 survival model architectures for a total of 15 combinations of survival models.

The RSF combined model was trained on 70 genes based on RNA-seq, 4 genes based on SCNA, and 21 clinical covariates. The 95 RSF model features are listed in **Supplementary Table 1** in the Appendix.

Risk scores were calculated for each patient using the cumulative hazard function from the RSF model. Patients in the CPTAC validation cohort were stratified into higher-risk and lower-risk groups using a pre-specified cutoff defined as the median predicted risk score from the TCGA training set, ensuring no information from the validation cohort influenced the stratification threshold. Based on this, 27 patients were flagged as higher-risk and 77 patients were flagged as lower-risk in the CPTAC cohort (**Figure 2**). Then, Kaplan-Meier survival

analysis using the log-rank test demonstrated a statistically significant separation between risk groups (log-rank p-value = 1.3×10^{-5} ; **Figure 2**), with the higher-risk group exhibiting worse overall survival.

Figure 2: RSF risk stratification (CPTAC validation)

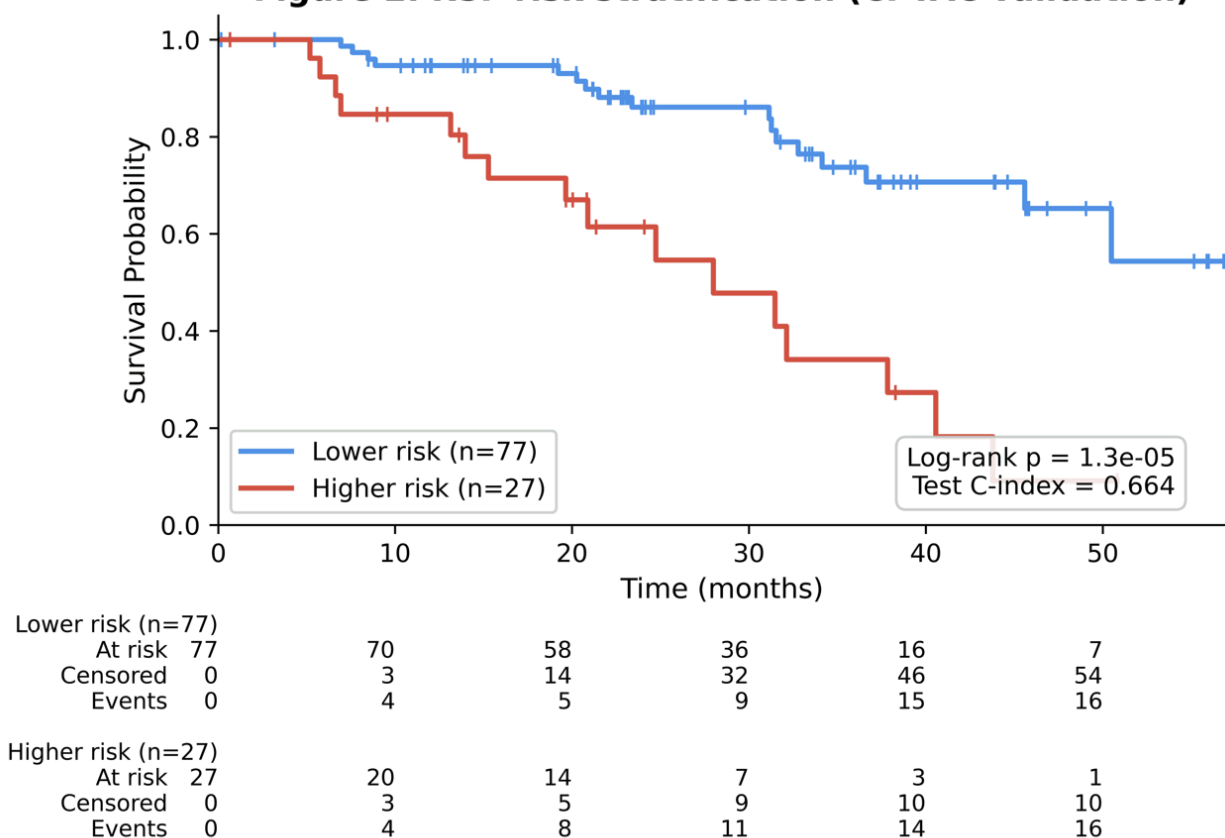


Figure 2: Kaplan-Meier survival curves for RSF risk stratification on the CPTAC validation cohort. Patients ($n=104$) were stratified into higher-risk ($n=27$) and lower-risk ($n=77$) groups using the median predicted risk score from the TCGA training cohort as the cutoff. Survival distributions differed significantly between groups (log-rank $p = 1.3 \times 10^{-5}$), with 16 events in each group (59.3% vs. 20.8% event rate). Test C-index = 0.664. Tick marks indicate censored observations.

Finally, over-representation analysis of the 74-gene signature revealed significant enrichment for serine-type endopeptidase activity (GO Molecular Function; FDR adjusted p-value = 0.046), driven by genes such as CTSG, PLAUI, MASP1, and TPSAB1 (**Figure 3**). At a less stringent threshold (FDR adjusted p-value ≤ 0.25), additional pathway enrichments were observed for extracellular matrix organization (Reactome; FDR = 0.241; 5 genes), collagen biosynthesis and modifying enzymes (Reactome; FDR = 0.241; 3 genes), activation of matrix metalloproteinases (Reactome; FDR = 0.241; 2 genes), and plasminogen activation (GO BP; FDR = 0.221) (**Figure 3**). These findings suggest that the prognostic signature captures a serine protease-mediated ECM-remodeling axis within the tumor microenvironment which influences overall survival.

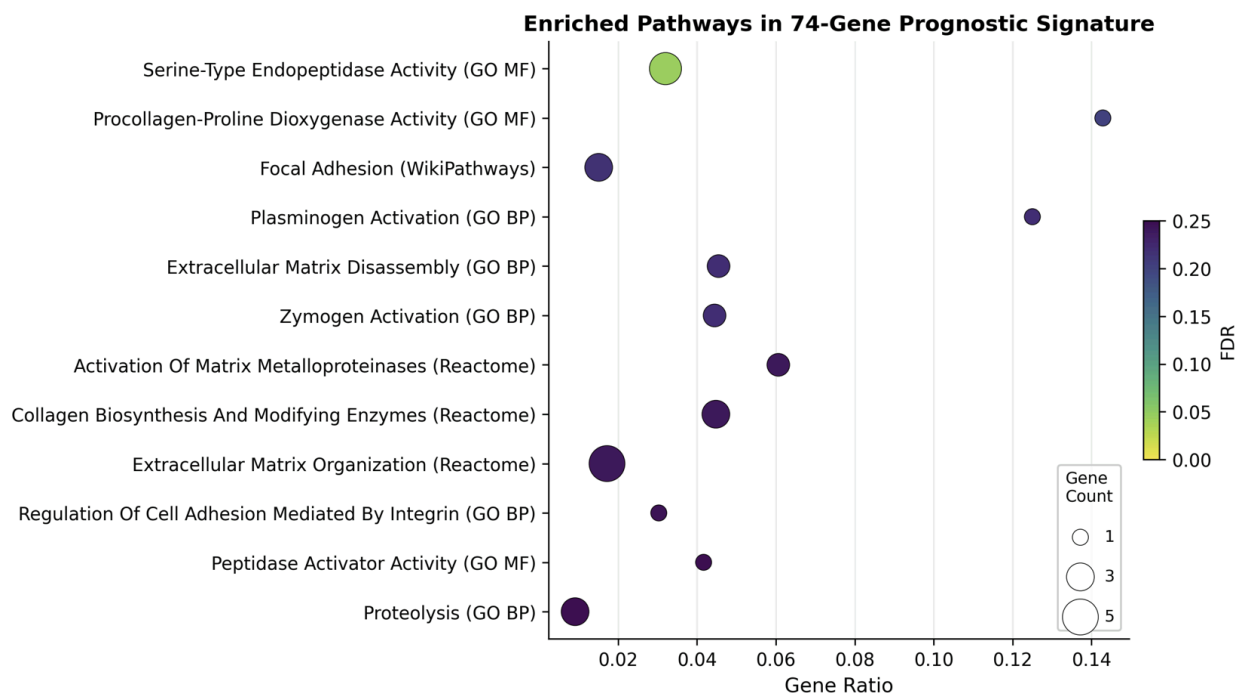


Figure 3. Pathway enrichment analysis of the 74-gene prognostic signature (70 RNA + 4 SCNA). Over-representation analysis was performed using Enrichr across GO, Reactome, and WikiPathways databases ($FDR \leq 0.25$). Dot size indicates gene count, color indicates FDR-adjusted p-value, and x-axis shows gene ratio.

Discussion and Conclusions

This project develops and validates a multi-omic prognostic model for overall survival in HPV-negative head and neck squamous cell carcinoma (HNSCC) by integrating RNA-seq data, somatic copy number alterations (SCNA), and clinical features. Across the survival modeling approaches evaluated, the Random Survival Forest (RSF) architecture achieved the strongest performance on the independent CPTAC cohort, with a fairly high test concordance index (C-index) of 0.664. This improved performance on an external cohort supports the utility of combining multi-omics and clinical information to improve survival prediction beyond what any single modality can provide.

The strength of the RSF model is in part due to the aggressive feature selection employed during training, which minimizes model overfitting. Indeed, the RSF model exhibited the lowest drop-off in C-index between the training and test set. Additionally, RSF models are well-suited to survival prediction in complex diseases because they can learn nonlinear relationships between clinical variables and molecular features, which may be difficult for largely linear models, such as CoxNet, to capture. Consistent with this, the combined RSF model outperformed the corresponding clinical-only and omics-only variants on the external validation cohort, indicating that the integration of clinical context with tumor genomics improves the prediction of patient outcomes.

Additionally, the RSF model demonstrated the ability to stratify patients from the test set into lower-risk and higher-risk categories, and these two groups had a statistically significant

difference in overall survival. Lower-risk patients could be recommended for primary surgery or radiation therapy alone, whereas patients identified as higher risk could be targeted with more aggressive treatment strategies, such as EGFR inhibitors and high-dose cisplatin-based chemoradiotherapy.

The prognostic signature includes 70 RNA-seq genes and 4 SCNA genes, and pathway analysis revealed enrichments for serine protease (endopeptidase) activity and extracellular matrix (ECM) remodeling. The enrichment is driven by proteases such as Cathepsin G (encoded by CTSG) and Urokinase-type plasminogen activator (encoded by PLAU), which participate in ECM breakdown and plasminogen activation in the tumor microenvironment.^{37,38} Furthermore, these enzymes serve to activate matrix metalloproteinases, further facilitating the remodeling of the ECM. Notably, the identified serine proteases are predominantly extracellular or cell-surface-associated, improving their pharmacologic accessibility. This, along with the demonstrated importance of these proteins in driving tumor progression indicates that they could serve as crucial targets for drugs such as upamostat and gabexate.^{39,40}

A key strength of this project is the use of an external cohort to assess model performance. While some papers report higher C-indices, they often do not assess model performance on an external cohort, and as a result, these models are commonly overfitted on a training set, worsening their ability to generalize to novel contexts. Another strength is restricting the samples to only HPV-negative HNSCC, as many previous survival model papers do not stratify by HPV status, which can be problematic given that HPV-negative and HPV-positive HNSCC are two molecularly distinct diseases.¹ For this reason, models that do not separate HPV-positive and HPV-negative HNSCC samples could just be learning to differentiate between the two, inflating c-indices. Despite its strengths, a clear limitation of this project is the small validation cohort size ($n = 104$) which reduces the precision of survival model performance estimates.

Therefore, future work should increase the sizes of the training and test sets, given that machine learning models become more accurate as the size of the training data grows.⁴¹ In addition, scaling the pipeline of this project to incorporate even more modalities would likely improve model performance given that tumors are rewired at every -omics layer. Finally, the genes highlighted in the prognostic signature warrant further pharmacological and statistical investigation, including whether the serine protease/ECM remodeling signal is reproducible across datasets and whether it associates with known tumor microenvironment phenotypes or treatment response patterns.

In conclusion, this project demonstrates that integrating RNA-seq, SCNA, and clinical features enables externally validated survival prediction in HPV-negative HNSCC, with the RSF model achieving a test C-index of 0.664 on an independent cohort and producing meaningful risk stratification. These results support multi-omic integration as a promising strategy for improving survival prediction models in both HPV-negative HNSCC and in other cancer subtypes. These approaches hold great potential, and could eventually be translated into the clinic, where higher-risk patients denoted by specific molecular markers could receive more aggressive treatment regimens.

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Appendix

Supplementary Table 1: List of features in the final random survival forests model to improve the transparency and reproducibility of the project findings

Feature #	Feature Name	Feature Type	Description
1	Age	Clinical	Patient age at diagnosis (years)
2	Pack_Years	Clinical	Cumulative tobacco exposure (pack-years)
3	Gender_Male	Clinical	Male sex (vs Female reference)
4	Stage_II	Clinical	Tumor stage II (vs Stage I reference)
5	Stage_III	Clinical	Tumor stage III (vs Stage I reference)
6	Stage_IV	Clinical	Tumor stage IV (vs Stage I reference)
7	Stage_Unknown	Clinical	Tumor stage unknown (vs Stage I reference)
8	T_Stage_T1	Clinical	Primary tumor T1 (vs T0 reference)
9	T_Stage_T2	Clinical	Primary tumor T2 (vs T0 reference)
10	T_Stage_T3	Clinical	Primary tumor T3 (vs T0 reference)
11	T_Stage_T4	Clinical	Primary tumor T4 (vs T0 reference)
12	T_Stage_TX	Clinical	Primary tumor TX (vs T0 reference)
13	N_Stage_N1	Clinical	Lymph node stage N1 (vs N0 reference)
14	N_Stage_N2	Clinical	Lymph node stage N2 (vs N0 reference)
15	N_Stage_N3	Clinical	Lymph node stage N3 (vs N0 reference)
16	N_Stage_NX	Clinical	Lymph node stage NX (vs N0 reference)
17	Grade_G2	Clinical	Moderately differentiated (vs G1 reference)
18	Grade_G3	Clinical	Poorly differentiated (vs G1 reference)
19	Grade_GX	Clinical	Grade unknown (vs G1 reference)
20	Alcohol_History_Unknown	Clinical	Alcohol history unknown (vs No reference)
21	Alcohol_History_Yes	Clinical	History of alcohol consumption (vs No reference)
22	OAF	Transcriptomic	OAF gene expression
23	ZNF662	Transcriptomic	ZNF662 gene expression
24	CPNE5	Transcriptomic	CPNE5 gene expression
25	FLT3	Transcriptomic	FLT3 gene expression
26	GLP1R	Transcriptomic	GLP1R gene expression
27	RTTN	Transcriptomic	RTTN gene expression

28	TPSAB1	Transcriptomic	TPSAB1 gene expression
29	EVA1C	Transcriptomic	EVA1C gene expression
30	SLC12A3	Transcriptomic	SLC12A3 gene expression
31	JCHAIN	Transcriptomic	JCHAIN gene expression
32	PCOLCE2	Transcriptomic	PCOLCE2 gene expression
33	SPINK6	Transcriptomic	SPINK6 gene expression
34	TOR1B	Transcriptomic	TOR1B gene expression
35	PLAU	Transcriptomic	PLAU gene expression
36	BCO1	Transcriptomic	BCO1 gene expression
37	ZNF557	Transcriptomic	ZNF557 gene expression
38	PTGDS	Transcriptomic	PTGDS gene expression
39	ATG5	Transcriptomic	ATG5 gene expression
40	ZAN	Transcriptomic	ZAN gene expression
41	B4GALNT3	Transcriptomic	B4GALNT3 gene expression
42	SPDL1	Transcriptomic	SPDL1 gene expression
43	FGF22	Transcriptomic	FGF22 gene expression
44	CALML5	Transcriptomic	CALML5 gene expression
45	EXPH5	Transcriptomic	EXPH5 gene expression
46	CTNNB1	Transcriptomic	CTNNB1 gene expression
47	ZNF44	Transcriptomic	ZNF44 gene expression
48	PLGLB1	Transcriptomic	PLGLB1 gene expression
49	AMN	Transcriptomic	AMN gene expression
50	CTSG	Transcriptomic	CTSG gene expression
51	MS4A1	Transcriptomic	MS4A1 gene expression
52	TPPP3	Transcriptomic	TPPP3 gene expression
53	CARMIL3	Transcriptomic	CARMIL3 gene expression
54	MAPK9	Transcriptomic	MAPK9 gene expression
55	EVPLL	Transcriptomic	EVPLL gene expression
56	GPN3	Transcriptomic	GPN3 gene expression
57	TNFRSF4	Transcriptomic	TNFRSF4 gene expression
58	PTX3	Transcriptomic	PTX3 gene expression
59	CCR7	Transcriptomic	CCR7 gene expression
60	HDC	Transcriptomic	HDC gene expression
61	PTPRZ1	Transcriptomic	PTPRZ1 gene expression
62	CFL2	Transcriptomic	CFL2 gene expression

63	PTPRR	Transcriptomic	PTPRR gene expression
64	TCL1A	Transcriptomic	TCL1A gene expression
65	TCP11	Transcriptomic	TCP11 gene expression
66	PCMT1	Transcriptomic	PCMT1 gene expression
67	FRMD5	Transcriptomic	FRMD5 gene expression
68	PTX4	Transcriptomic	PTX4 gene expression
69	COL6A5	Transcriptomic	COL6A5 gene expression
70	S1PR4	Transcriptomic	S1PR4 gene expression
71	ZNF266	Transcriptomic	ZNF266 gene expression
72	ANO2	Transcriptomic	ANO2 gene expression
73	CCL20	Transcriptomic	CCL20 gene expression
74	ANKMY1	Transcriptomic	ANKMY1 gene expression
75	SURF4	Transcriptomic	SURF4 gene expression
76	DNAJB13	Transcriptomic	DNAJB13 gene expression
77	RAB44	Transcriptomic	RAB44 gene expression
78	RBM18	Transcriptomic	RBM18 gene expression
79	P4HA1	Transcriptomic	P4HA1 gene expression
80	CELSR3	Transcriptomic	CELSR3 gene expression
81	PPID	Transcriptomic	PPID gene expression
82	AIG1	Transcriptomic	AIG1 gene expression
83	NSMCE3	Transcriptomic	NSMCE3 gene expression
84	SLC23A1	Transcriptomic	SLC23A1 gene expression
85	BLK	Transcriptomic	BLK gene expression
86	CCDC88B	Transcriptomic	CCDC88B gene expression
87	EML6	Transcriptomic	EML6 gene expression
88	CCDC32	Transcriptomic	CCDC32 gene expression
89	MASP1	Transcriptomic	MASP1 gene expression
90	SLC25A45	Transcriptomic	SLC25A45 gene expression
91	MYO1H	Transcriptomic	MYO1H gene expression
92	XAB2	Copy Number	XAB2 copy number alteration
93	ZNF479	Copy Number	ZNF479 copy number alteration
94	MRPS17	Copy Number	MRPS17 copy number alteration
95	ZNF713	Copy Number	ZNF713 copy number alteration